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ARTIFICIAL INTELLIGENCE (19242 21134)

BTECH-IT

CO1

Assessment-3

1. * A Smart home automation System is an intelligent System that automatically controls lighting, temperature and Security by analyzing real time inputs, environmental conditions and user preferences, It uses Sensors, actuators, and intelligent agents to make decisions with minimal human intervention.

Working principle :

Sensors and User inputs

↓

Intelligent Agent

↓

Actuators and Actions

(i) Simple Reflex agent

(ii) Model Based Reflex agent

(iii) Goal Based Agent

(iv) Utility Based Agent

ADVANTAGES

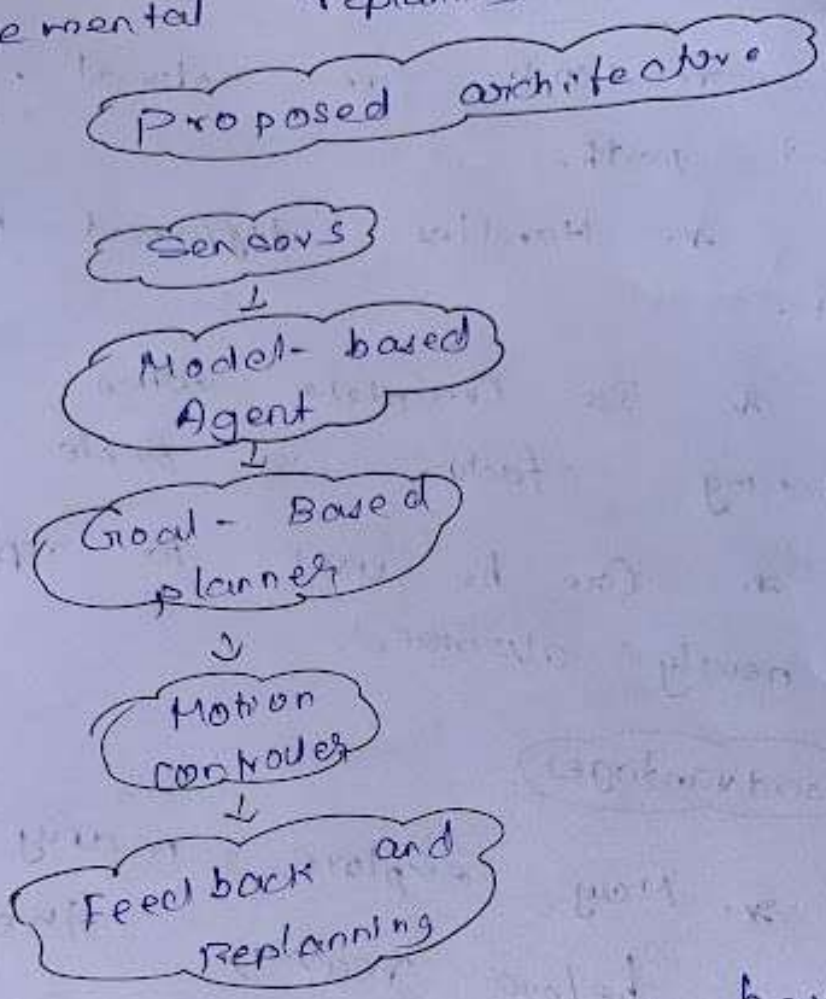
* Finds an optimal, minimal cost path.
* Handles different movement cost.
* Is complete when the branching factor is finite
* Can be used to replan when newly discovered.

Disadvantages

* May explore many positions before reaching the exit.
* Stores frontier nodes and explored state - requiring substantial memory

1.) Model - based design
2.) Goal - based decision
3.) Uniform cost search
4.) Reflex safety behavior

Incremental replanning

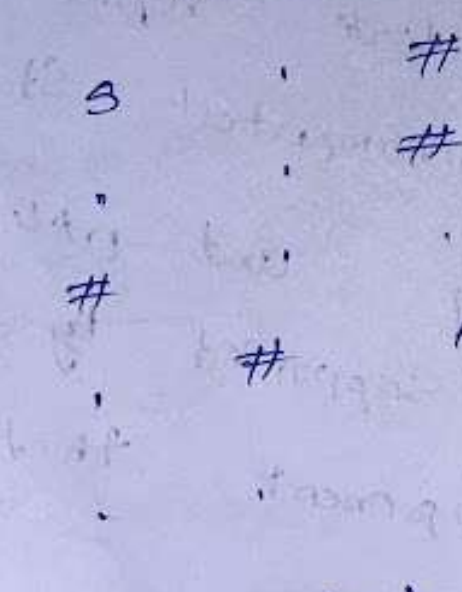


*. A model- based goal- based agent Combined with UCS is generally the strongest choice when route cost matters and sufficient memory is available.

It remembers the environment, systematically Searches for exit and adapts as new information becomes available.

(2.) * A robot placed in an unknown maze must find a path from its starting position to the exit without knowing the maze layout in advance. It uses sensors to detect walls, open paths and possible movements. By exploring the environment and recording discovered locations the robot can search for a route to exit.

S = Start, E = Exit, # = Wall



Uniform cost search UCS

$$g(n) = \sum_{i=1}^{K} C_i$$

*. Safe operation.

*. Evaluation

Intelligent agents enable a smart home to react to environmental conditions and make increasingly sophisticated decisions. Simple reflex agents provide fast responses, model-based agents plan toward desired states. and utility-based agents balance competing objectives

therefore a layered architecture is the most effective choice. It combines immediate reflex actions, goal-directed planning and utility-based optimization. Supported by a learning component that adapts to user preferences

Application in Smart homes

* Lighting
* Temperature
* Security
* Learning preferences

* The most effective solution is a hybrid intelligent - agent architecture combining model - based, goal - based, with simple reflex rules for urgent responses.

Level 1: Simple Reflex Agent

Layer 2: Model Based Agent

Layer 3: Goal - Based Agent

Layer 4: Utility Based Agent

Challenges and Safeguards

* Sensor reliability
* privacy
* Security priority
* User control